Floor Area and Regional Differences in New Home Prices

STA302H1 Fall 2026 | Group 62 | Part 1 Research Proposal

Contributions

Group 62

Member name	Contribution to this proposal
Haiwen Yi	Research design, statistical analysis, and report writing.
Olivia Qi	Revision-result interpretation, details, structure and format review
Matthew Frieri	Revised proposal against rubric, bringing sections up to par with criterion
Annie Xu	Independent Rmd verification, data and numerical consistency checks

Introduction

Among new detached, non-condominium homes sold in the United States in 2025, how is sale price (including land, in thousands of dollars) associated with finished floor area (in hundreds of square feet), and does this association differ across the nine Census divisions (regional groupings of states) after adjustment for lot size, room counts, amenities, construction features and sale timing?

Prior evidence gives two reasons to question a uniform area–price relationship. In a meta-regression of U.S. single-family housing studies, Sirmans et al. (2006) found geographic differences in floor-area coefficients and reported that hedonic model size affected those coefficients. This motivates comparing regions using the same adjustment set. A possible economic mechanism is variation in building costs: across U.S. housing markets, Gyourko and Saiz (2006) found substantial cost differences associated with wages, unionization, terrain and regulation. Because our response includes both structure and land, however, a regional area coefficient cannot be interpreted as construction cost. Heterogeneity also occurs within markets: using over 136,000 single-family sales in Jacksonville, Florida, during 1990–2006, Zietz et al. (2008) found larger implicit prices for floor area and lot size at higher price quantiles. Thus, geographic and price-segment heterogeneity are distinct; their quantile results do not establish differences between regional mean slopes.

Our contribution is a common-data comparison for 2025 new detached homes: we estimate regional conditional mean slopes under one population definition, year and specification, rather than compare coefficients assembled from different studies. Adjusting for lot size and room counts makes the area coefficient describe differences in interior space at otherwise fixed recorded characteristics. The analysis tests regional variation but cannot identify whether costs, preferences or unmeasured location explain it.

Regional averages and pairwise correlations cannot provide these conditional comparisons. Multiple linear regression includes floor area, division and their interaction alongside the adjustment variables. Nine estimated slopes and confidence intervals quantify price differences associated with 100 additional square feet; a joint test of the eight interaction coefficients assesses whether the slopes differ. Builders comparing floor plans and appraisers selecting size adjustments could use regional slopes as market context, checking whether a national adjustment conceals differences. These are neither individual valuations nor causal returns to expansion. The linear form requires diagnostic checking, and unweighted complete-case analysis limits population generalization.

Data description

The 2025 Survey of Construction (SOC) microdata record sampled houses (U.S. Census Bureau, 2026). SOC produces statistics on housing starts, completions and sales using sampled permits, canvassing and builder or owner interviews (U.S. Census Bureau, n.d.-a).

Of 25,246 houses, 7,382 are eligible. Excluding unreported price (320), area or lot (747), then missing categories (130) leaves 6,185 complete cases (4,330 training; 1,855 test); 16.2% exclusion risks bias.

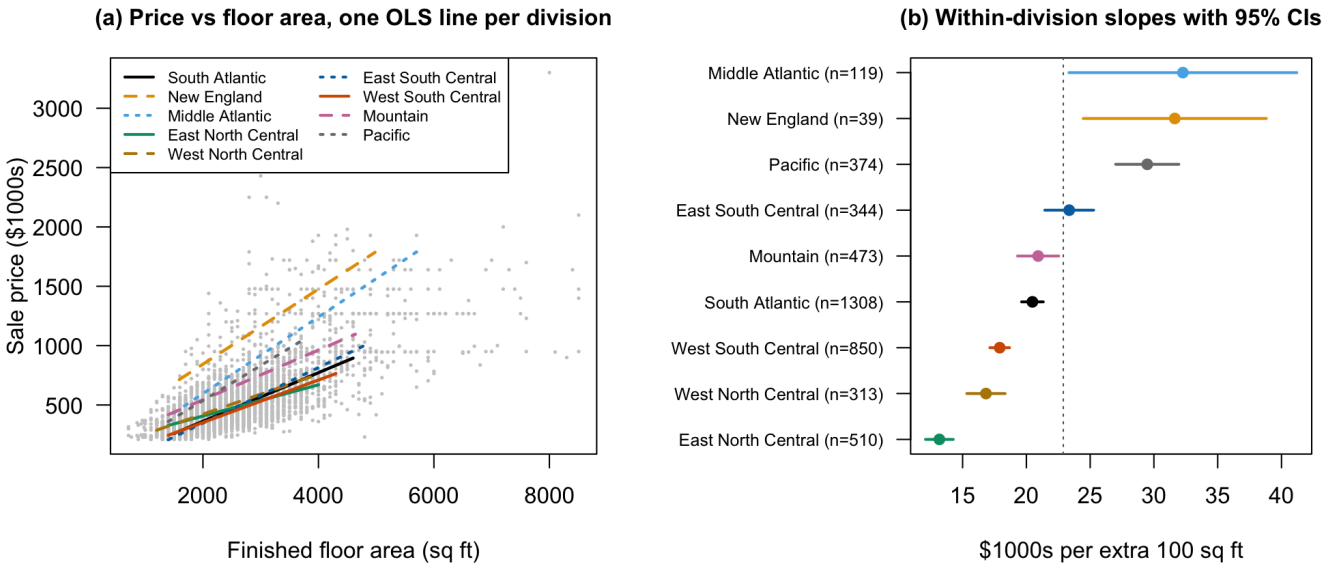
Approximately continuous price (Table 1) supports conditional-mean regression; shared builders may threaten independence. Area–division interaction tests geographic slope differences (Sirmans et al., 2006), retaining native divisions; see Figure 1.

Table 1. *Training sample summaries.*

Variable	Summary
Price ($1,000)	527 (307); median: 433; range: 210–3,300
Area (100 sq ft)	25.4 (9.9)
Lot (1,000 sq ft)	14.1 (16.6)
Sale month	6.3 (3.4)
Division	South Atlantic: 1,308; New England: 39; Middle Atlantic: 119; East North Central: 510; West North Central: 313; East South Central: 344; West South Central: 850; Mountain: 473; Pacific: 374
Stories	1: 2,120; 2: 2,116; 3+: 94
Bedrooms	≤2: 154; 3: 1,780; 4: 1,777; 5+: 619
Full bathrooms	≤1: 46; 2: 2,664; 3: 1,203; 4+: 417
Half bathrooms	0: 2,373; 1: 1,897; 2+: 60
Garage	1-car: 158; 2-car: 3,124; 3+-car: 919; other: 129
Foundation	Basement: 992; crawl space: 403; slab: 2,926; other: 9
Fireplaces	0: 2,781; 1: 1,383; 2+: 166
HOA	No: 1,109; yes: 3,221
Age restriction	No: 4,216; yes: 114
Wall	Brick: 897; wood: 169; stucco: 906; vinyl: 1,362; stone: 56; fiber cement: 926; other: 14
Heating	Electric: 2,147; gas: 2,083; propane: 52; other: 48
Sewer	Public: 3,893; septic/other: 437
Sale stage	Completed: 2,452; under construction: 1,344; not started: 534

Note. Means (standard deviations) or counts; reference categories first. Price, area and lot are right-skewed; extremes disclosure-coded (U.S. Census Bureau, n.d.-b).

Figure 1. *Unadjusted price–area relationships; vertical line: unweighted mean of division slopes.*



Ethics discussion

The first ethics module asks whether data are collected or simulated, whether metadata and original sources are documented, and whether third-party use supports trustworthiness. SOC records come from a documented Census survey, not simulated examples; the published definitions, sampling information and disclosure flags provide traceable metadata. On third-party use, SOC data underlie official federal economic indicators and are used by HUD, the Bureau of Economic Analysis, the National Association of Home Builders and Harvard's Joint Center for Housing Studies. Government origin supports provenance but does not remove sampling, reporting or measurement error.

For privacy, the public file suppresses detailed geographic information, removes addresses and other identifiers, and codes extreme values (U.S. Census Bureau, n.d.-b). We will not attempt re-identification or publish individual-house appraisals. Regarding consent, public statistical release is distinct from participants consenting to arbitrary new uses: we limit our use to aggregate housing research compatible with the collection purpose and do not claim to have independently audited every respondent's consent. Regarding ownership, we acknowledge the Census Bureau and HUD and retain source attribution. The estimates describe conditional associations; regional labels must not be used to stereotype households or imply causal effects of adding floor space.

Preliminary results

Unweighted OLS uses 17 predictors and the area–division interaction (52 coefficients; n=4,330). Training R^2=0.713 (adjusted R^2=0.710); the interaction adds 0.020. A partial F-test of the eight interaction terms gives $F(8,4278)=37.27$, $p<0.001$, indicating slope differences if OLS assumptions hold.

Holding other predictors fixed (Table 2), 100 additional square feet corresponds to $11,834 higher mean price in South Atlantic and $24,557 in Middle Atlantic: a $12,723 slope difference (95% CI: $10,359–$15,087). New England's n=39 limits precision.

Condition 1 (conditional mean response) seems to hold: Figure 2A shows a simple increasing curve despite upper-range departures from the identity line. Condition 2 (conditional mean predictor) seems to hold for the plotted numerical pairs: 2B bends mildly; 2C–D are nearly flat. These checks support interpreting residual plots. Linearity (mean-zero errors) appears mildly violated: smoothers curve (Figures 3A, 3D); lot and month smoothers remain near zero (3E–F). Constant error variance appears violated: fanning and rising scale–location (3A–B), with differing group spreads (Figures 4A–E, 5A); regional residual SDs differ fivefold. Normality appears violated: heavy Q–Q tails, especially right (3C); 36 standardized residuals lie beyond ±4 (4F). Uncorrelated errors remain unverified: the month smoother (3F) cannot rule out dependence; builder identifiers are unavailable. One violation may affect another assumption's diagnostic plot. Inference remains provisional; no corrections applied.

Regional slope differences are unsurprising given Sirmans et al. (2006), but magnitudes are not directly comparable across populations and specifications. Construction costs remain a possible, untested explanation (Gyourko & Saiz, 2006), not an identified cause.

Table 2. *Adjusted floor area slopes by Census division ($1,000 per 100 sq ft; South Atlantic reference).*

Division	n	Slope	SE	95% CI	Difference from South Atlantic (SE)
South Atlantic (reference)	1,308	11.83	0.63	(10.60, 13.07)	—
New England	39	19.26	2.15	(15.05, 23.47)	7.43 (2.17)
Middle Atlantic	119	24.56	1.14	(22.32, 26.79)	12.72 (1.21)
East North Central	510	6.85	0.85	(5.19, 8.51)	−4.98 (0.91)
West North Central	313	10.57	1.15	(8.32, 12.82)	−1.26 (1.20)
East South Central	344	13.78	0.97	(11.88, 15.68)	1.95 (1.00)
West South Central	850	9.27	0.79	(7.71, 10.82)	−2.57 (0.82)
Mountain	473	14.60	0.80	(13.03, 16.17)	2.77 (0.87)
Pacific	374	21.70	1.14	(19.47, 23.92)	9.86 (1.17)

Figure 2. *Conditional-mean response and predictor checks.*

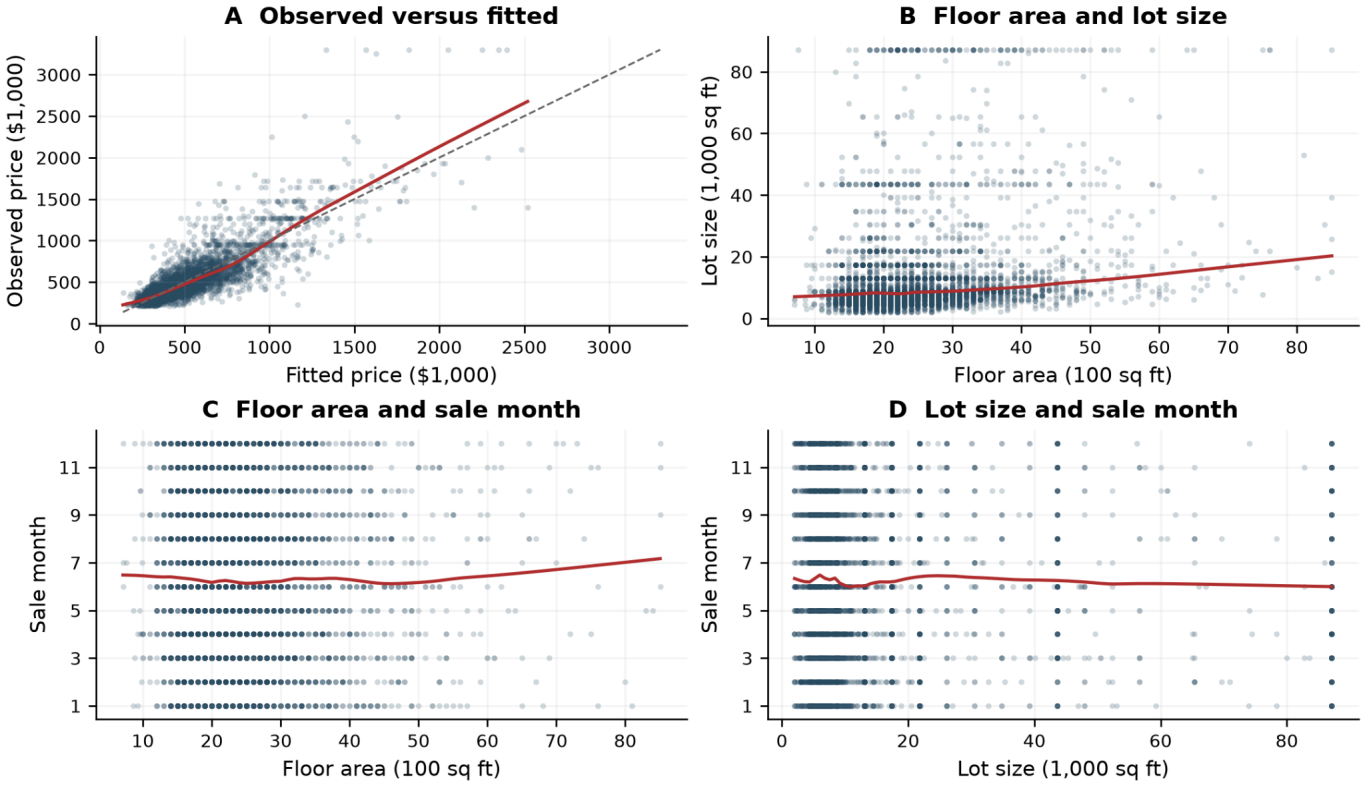


Figure 3. *Residual diagnostics for the preliminary regression model.*

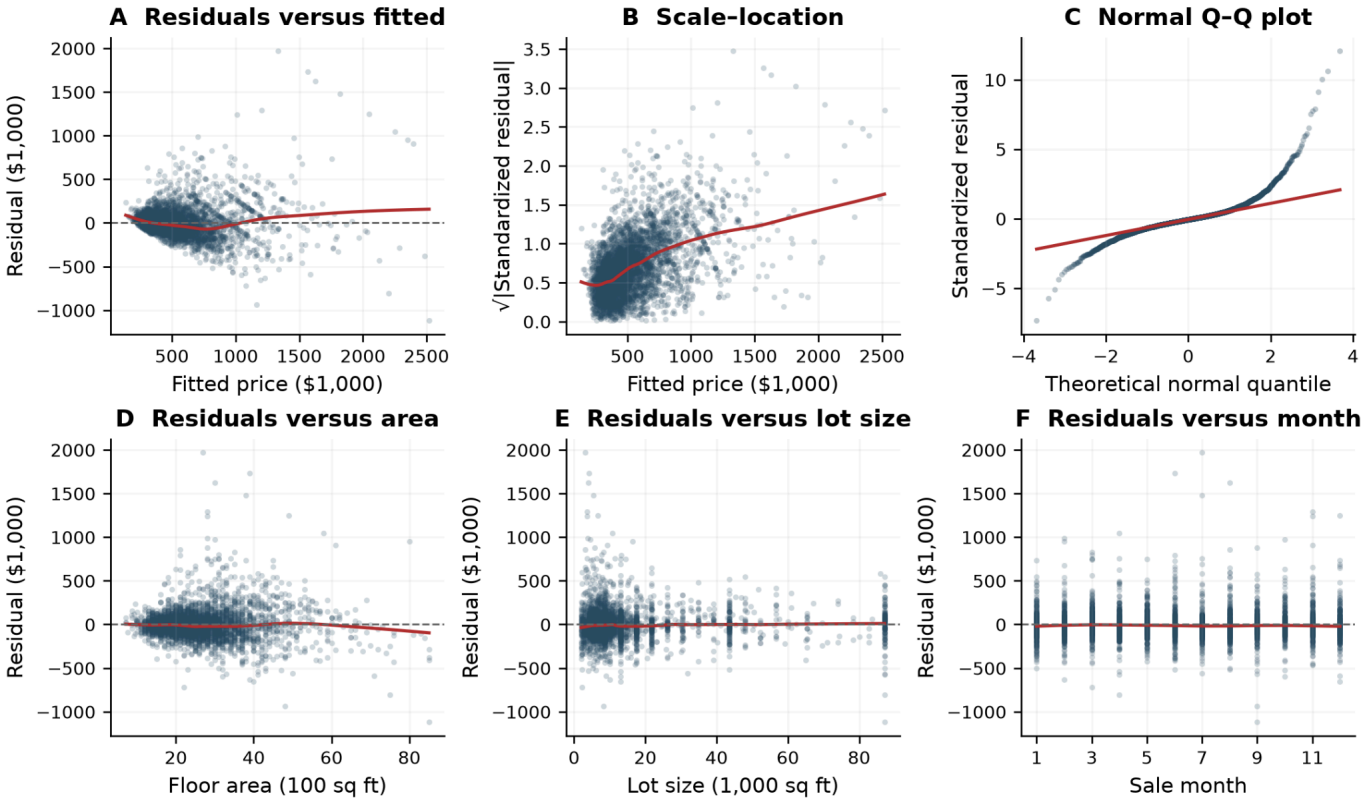


Figure 4. *Selected categorical residual distributions and standardized-residual histogram.* SD ratios compare largest/smallest within-group SDs; boxplots omit outlier points.

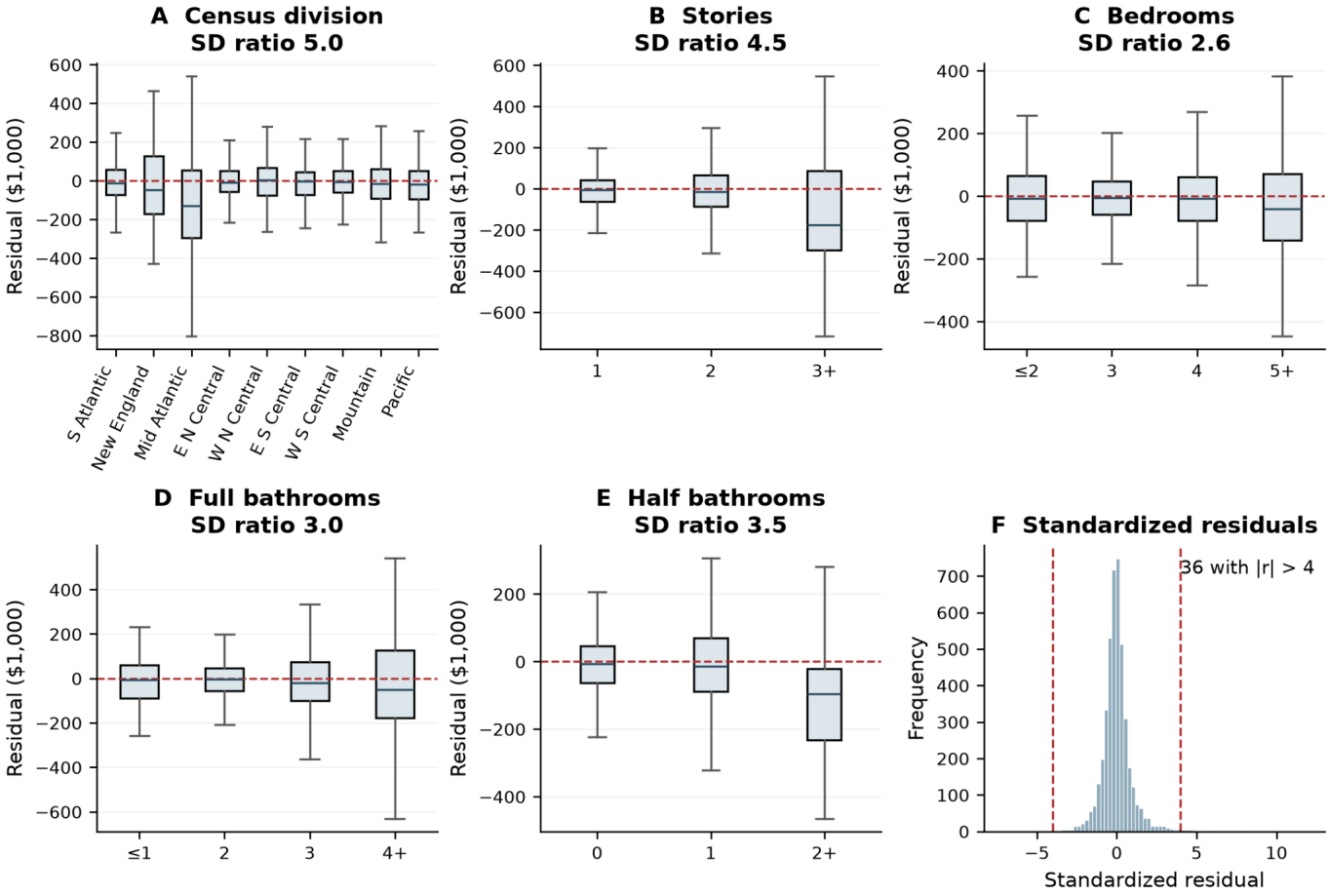
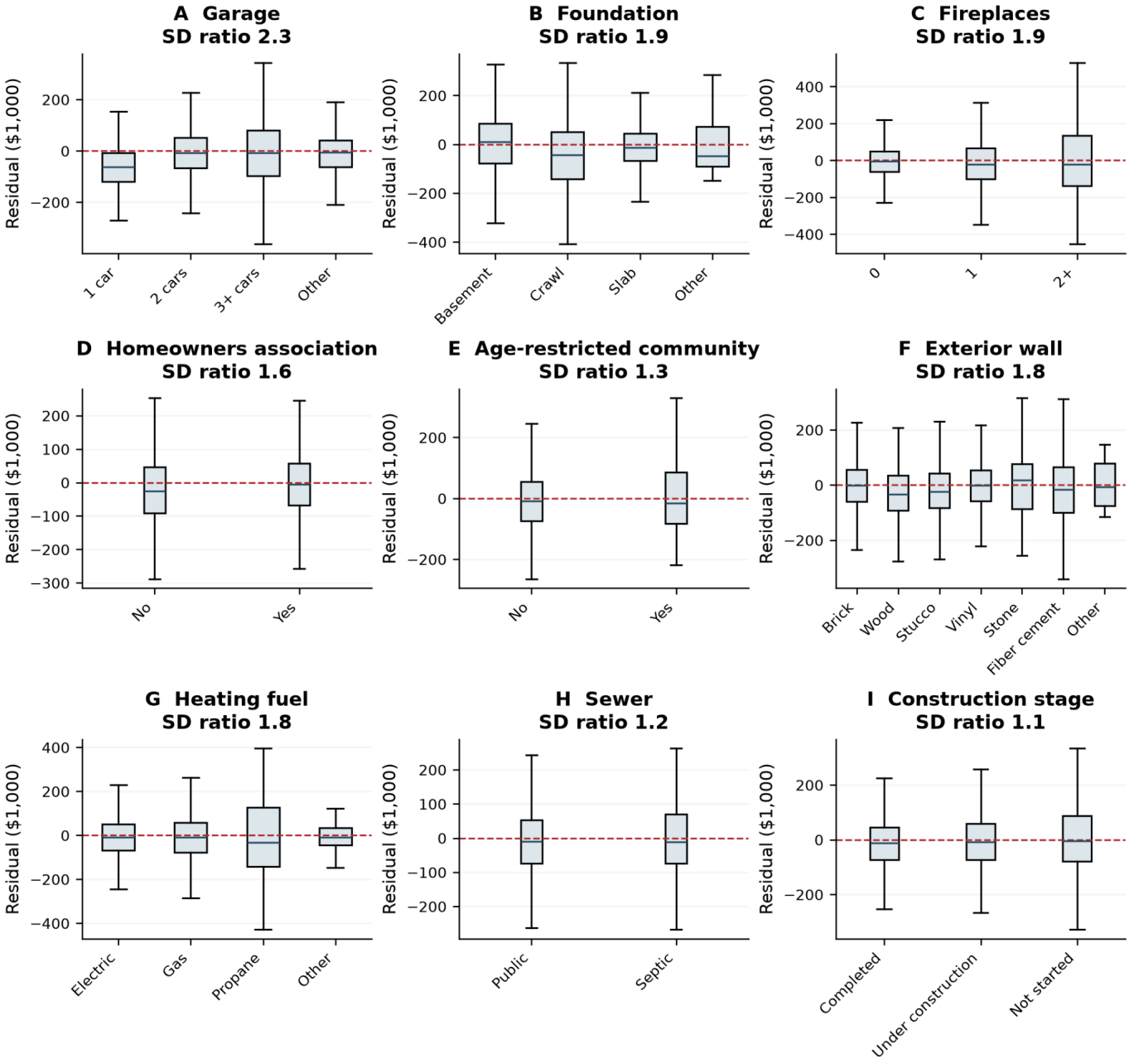


Figure 5. *Additional categorical residual distributions; outlier points hidden.*



Plan

Retain floor area, division and their interaction for the research question; retain lot size because Zietz et al. (2008) found it priced alongside floor area, consistent with hedonic theory (Rosen, 1974). Select other predictor groups by backward BIC, preserving interaction hierarchy. Use nested F-tests as supporting evidence and AIC for sensitivity checks on area slopes, since hedonic model size affected area coefficients in Sirmans et al. (2006).

Before predictor selection, assess Box–Cox price transformations, including log price, for fanning and heavy tails; if curvature remains, compare log-area or quadratic-area terms. Log-price slopes describe proportional rather than dollar differences. Recheck Conditions 1–2 and residual diagnostics after every change; transformation may not resolve misspecification.

After improving the mean structure, use HC3 inference if unequal variance persists; consider weighted least squares only with a defensible variance model. Assess influential cases and survey-weight sensitivity; remove observations only for documented errors. Unidentified clustering remains a limitation that HC3 cannot fix.

Select transformations using training diagnostics, then freeze the model before one held-out test evaluation.

Table 3. *Team schedule.*

Date	**Deliverable**	**Lead**
Nov 5	Submit Parts 0–1.	All
Before final submission	Hold at least two TA/professor consultations for rubric-based feedback; document suggestions, revise the project and verify improvements.	Coordinator: TBD
Nov 19	Compare models; assess influential cases.	Haiwen Yi
Nov 22	Complete robustness checks; freeze model; evaluate test set.	Haiwen Yi
Nov 27	Draft poster content, figures and layout.	Olivia Qi
Nov 27	Complete presentation script and visual design.	TBD
Nov 30	Complete poster, rehearsal and recording.	All
Dec 1	Check rubric, numbers, citations and recording; return corrections.	Annie Xu
Dec 2	Resolve feedback; finalize files and individual reflections.	All
Dec 3	Submit Parts 2–3; verify receipts.	All

Internal targets except submissions (23:59 Toronto time). Consultations subject to availability; TBD roles await volunteers.

Bibliography

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